

# InSciTe Capstone Project Log

This log will be used to track the progress, challenges, and decisions made throughout the capstone project.

## Project Information

| Field                    | Detail                        |
|--------------------------|-------------------------------|
| Project Title            | Character-driven AI Assistant |
| Team Members             | Gerrion Purnell               |
| Faculty Advisor          | Dr. Kinnicutt                 |
| Start Date               | Aug 29, 2025                  |
| Expected Completion Date | May 2, 2026                   |

## Log Entries

Week 1-3: Aug 31, 2025 - Sep 20, 2025

- **Goals for Next Three Weeks:**
  - Create a basic chatbot framework using if-then statements.
  - Work on and refine project pitch.
  - Integrate basic memory, mood, and personality into the chatbot
- **Progress Made:**
  - Basic chatbot is functional.
  - Created and presented the project pitch.
  - Basic memory, mood, and personality is functional.
- **Challenges/Roadblocks:**
  - Chatbot is limited in understanding and response.
  - Looking for a small local LLM to be integrated into the program.
- **Solutions/Workarounds:**
  - Adding the LLM would fix the response limitation of the chatbot.
  - Looked into places like Ollama for local LLMs.
- **Decisions Made:**

- Decided not to focus on functionality until LLM is properly integrated and functioning.
- Decided not to focus too heavily on the simple chatbot and focus more on LLM integration.
- **Next Steps:**
  - Choosing and integrating a LLM into the program.
  - Testing various LLMs to see which local LLM runs best with the program.
  - Updating the personality system so it works with the LLM.

## Week 4-7: Sep 21, 2025 - Oct 18, 2025

- **Goals for the Next Three Weeks:**
  - Work on the project proposal.
  - Adding a LLM into the program
  - Adjust and test the personality system so it works with the LLM.
  - Testing various LLMs to see which one retains personality the best.
  - Improved the mood system so it works with LLM.
  - Set up github repository
- **Progress Made:**
  - Tested a wide variety of LLMs.
  - Added a LLM into the program.
  - Adjusted personality to work with LLM.
  - Made a context system so the LLM retains memory and personality better.
  - Started working on the project proposal.
  - Github repository is set up.
  - Finished half of the mood system.
- **Challenges/Roadblocks:**
  - Most small local LLMs don't have the power needed to utilize the personality or context.
  - Wasn't sure how to effectively apply mood so it works reliably when faced with unexpected input.
- **Solutions/Workarounds:**
  - Started looking into LLMs geared towards role playing to address the problem of LLM not retaining personality.
  - Used a sentiment system to give values to user response to determine what the mood should be.
- **Decisions Made:**
  - Decided to test with APIs as well as local LLMs
  - Decided to use VADER for sentiment analysis
- **Next Steps:**
  - Looking into fine-tuning a LLM
  - Finish mood system

- Start on refining memory systems

**Week 8-11:** Oct 19, 2025 - Nov 15, 2025

- **Goals for the Next Three Weeks:**

- Finishing mood system
- Refining personality
- Improving context
- Start working on memory
- Cleaning code by removing outdated code
- Make a separate actions class
- Clean Github repository

- **Progress Made:**

- Cleaned code
- Improved context
- Refined personality
- Started working on a separate actions class
- Finished mood system
- Started working on memory

- **Challenges/Roadblocks:**

- Sentiment doesn't effectively capture mood
- Sectioned parsing LLM relays unreliable information
- AI occasionally replies without personality
- Parsing LLM output affects whether or not the AI responds with personality
- Mood doesn't persist and doesn't stand out strongly
- Parsing LLM will create its own format for json output, which is incompatible with the current code causing it to crash

- **Solutions/Workarounds:**

- Letting the parsing LLM handle determining the mood
- Rearranging the order in which the LLM is given context and personality
- Rephrased text for parsing LLM so it compliments personality

- **Decisions Made:**

- Decided to split the response of the parsing AI and having it handle multiple tasks to decrease LLM calls
- Decided to make a separate class for actions
- Decided to store a user fact log in a file separate from other user information for easier access and recalling

- **Next Steps:**

- Finishing memory
- Making GUI
- Adding functionality

## Week 12-15: Nov 16, 2025 - Dec 13, 2025

- **Goals for the Next Three Weeks:**
  - Finish memory
- **Progress Made:**
  - Continued working on memory
- **Challenges/Roadblocks:**
  - The code I made to save the memory caused errors
- **Solutions/Workarounds:**
  - Started on re-working the code made to save memory
  - Re-thinking my approach to implementing memory
- **Decisions Made:**
  - Decided to use the winter break to re-think my approach to implementing memory
- **Next Steps:**
  - Finishing memory
  - Start working on GUI

## Week 16-19: Jan 11, 2026 - Feb 7, 2026

- **Goals for the Next Three Weeks:**
  - Finish memory
  - Start working on GUI
  - Work on refining memory
  - Fix memory duplication problem
  - Fix hallucination problem
  - Finish base GUI
  - Test code packaging
  - Fix json format problem
  - Start working of document reading
- **Progress Made:**
  - Finished base memory
  - Started working on GUI
  - Duplication problem fixed
  - Hallucination problem fixed
  - Finished base GUI
  - Json format problem fixed
- **Challenges/Roadblocks:**
  - Context format was causing errors when attempting to store memory
  - Multiple memory entries can appear for the same item
  - Parsing LLM occasionally mistakes the user for the AI
  - GUI freezes during AI response

- Making context more specific occasionally caused errors
- **Solutions/Workarounds:**
  - Made context format more specific
  - Adjusted the code that checks for duplicate entries in memory
  - Adjusted parsing LLM code to reduce the chances of hallucination
  - Added threads to the GUI
  - Reworked parsing LLM to improve context and remove format errors
- **Decisions Made:**
  - Decided to use CustomTkinter for GUI
  - Decided to add thinking animation
  - Decided to test packaging the code to make .exe file
  - Decided to let the reworked parsing LLM handle summarization
  - Decided to use PyPDF2 and python-docx for document reading
- **Next Steps:**
  - Continue working on GUI
  - Continue refining memory
  - Test and adjust AI
  - Look into confidence, stress, and trust value systems
  - Continue working on functionality and utility

Week 20-23: Feb 8, 2026 - Feb 28, 2026

- **Goals for the Next Three Weeks:**
  - Continue working on GUI
  - Continue refining memory
  - Test and adjust AI
  - Look into confidence, stress, and trust value systems
  - Continue working on functionality and utility
  - Add chunking for file information
  - Add embedding
- **Progress Made:**
  - AI can now relay weather information
  - Built a upload button
  - File chunking has been implemented
  - Embedding has been implemented
  - AI can now read files
  - Cleaned GUI
- **Challenges/Roadblocks:**
  - Wasn't sure how to properly use chunking in the code
- **Solutions/Workarounds:**
  - Learned how to use embedding to better deliver chunked information

- **Decisions Made:**
  - Decided to chunk file content
  - Decided to use embedding
  - Decided to make an upload button for files
- **Next Steps:**
  - Continue working on GUI
  - Continue refining memory
  - Test and adjust AI

## Week 24-27: Mar 1, 2026 - Mar 21, 2026

- **Goals for the Next Three Weeks:**
  - Continue working on GUI
  - Continue refining memory
  - Test and adjust AI
  - Add side menu for setting
- **Progress Made:**
  - Added side menu for settings
  - Added chat bubbles for GUI
  - Cleaned GUI
- **Challenges/Roadblocks:**
  - When the window is expanded the inputs and responses are too far apart
  - Large file uploads override personality when large amounts of content are needed to respond
- **Solutions/Workarounds:**
  - Made a smaller chat window inside the chat frame
- **Decisions Made:**
  - Decided to add settings menu
  - Decided to add chat bubbles
- **Next Steps:**
  - Continue working on GUI
  - Implement interchangeable personalities
  - Add personality menu

## Week 28-31: Mar 22, 2026 - Apr 11, 2026

- **Goals for the Next Three Weeks:**
  - Add personality menu
  - Implement interchangeable personalities
  - Continue working on GUI
  - Add setting persistence
  - Add settings reset button

- Add embedded memory
- Add short term memory
- **Progress Made:**
  - Added personality menu
  - Implemented interchangeable personalities
  - Added setting persistence
  - Added reset button for settings
  - Added embedded memory
  - Added short term memory
- **Challenges/Roadblocks:**
  - On rare occasions personalities other than Nyx will be unable to respond to queries involving information provided by external tools
  - Dropdown menu for personality wasn't properly updating when reset button was clicked
  - Embedded memory sometimes retrieved irrelevant memories
- **Solutions/Workarounds:**
  - Changed the code for the reset button to change the variable for the dropdown menu manually
  - Increased the minimum score for the embedded memory
- **Decisions Made:**
  - Decided to implement persistence for settings
- **Next Steps:**
  - Add memory reset button
  - Continue working on GUI
  - Add web searching

Week 32-35: Apr 12, 2026 - May 2, 2026

- **Goals for the Next Three Weeks:**
  - Add memory reset button
  - Continue working on GUI
  - Add web searching
  - Add weather switching
- **Progress Made:**
  - Fixed UI scaling issue
  - Fixed chat bubble spacing issue
  - Added memory reset button
- **Challenges/Roadblocks:**
  - Character AI hallucinates weather information when user asks for weather in places the weather api doesn't cover
  - AI can't help with some basic tasks without more information

- Embedding model sometimes uses an incompatible format
  - Shared memory sometimes causes personality bleed
- **Solutions/Workarounds:**
  - Allowed the AI to temporarily change the location to get weather information for other locations
  - Added a manual embedding fallback
- **Decisions Made:**
  - Decided to add the ability to change locations for weather API
  - Decided to allow the AI to temporarily change the location
  - Decided to make a separate embedding class
- **Next Steps:**
  - Continuing the development of the assistant independent of the university

## Resources

- Github: <https://github.io/Character-Driven-AI-Assistant-Demo/>